



DESCRIPTIVE STATISTICS PART 2

Session 39 — Bivariate analysis, covariance, correlation, and multi-variable visualization

Percentiles · Quartiles

Box Plot Deep Dive

Scatter Plots

Covariance

Pearson Correlation

Correlation $\neq$ Causation

Pair Plots · Heatmaps

Session Notes — Complete Reference

Playlist: Maths for Machine Learning · Duration: 1h 55m · Instructor: Nitish Singh (CampusX)



Table of Contents

- 01** Recap — Quantiles & Percentiles
- 02** Box Plot — Deep Dive & Variations
- 03** Introduction to Bivariate Analysis
- 04** Scatter Plots — Exploring Numerical-Numerical Relationships
- 05** Covariance — Direction of Joint Variability
- 06** Pearson Correlation Coefficient
- 07** Correlation vs Causation
- 08** Spurious Correlations & Confounding Variables
- 09** Simpson's Paradox
- 10** Bivariate Analysis — Numerical vs Categorical
- 11** Bivariate Analysis — Categorical vs Categorical
- 12** Multivariate Visualization — Pair Plots, Heatmaps & Beyond
- 13** Python Implementation — Complete Code
- 14** Quick Reference Card

1 • Recap — Quantiles & Percentiles

Before diving into bivariate analysis, this session begins with a recap of quantiles and percentiles from Part 1, solidifying these foundational concepts.

1.1 Quantiles — Generalised Concept

Quantiles are cut points dividing a probability distribution into equal-probability intervals. They are the generalisation of percentiles.

QUANTILE TYPE	NUMBER OF DIVISIONS	EXAMPLE
Quartiles	4 equal parts	Q1, Q2, Q3
Quintiles	5 equal parts	Used in finance (income brackets)
Deciles	10 equal parts	D1 through D9
Percentiles	100 equal parts	P1 through P99

1.2 Percentile Calculation Methods

There are multiple methods to calculate percentiles. The two most common:

Nearest-Rank Method

$$\text{Position} = \lceil (k/100) \times n \rceil$$

Take the value at the **ceiling** of the computed position. Always returns an actual data value.

Interpolation Method

$$\text{Position} = (k/100) \times (n + 1)$$

If position is not a whole number, **interpolate** between the two nearest values. More precise; used by NumPy/Pandas by default.

Example — 30th Percentile

Dataset (sorted): {2, 4, 6, 8, 10, 12, 14, 16, 18, 20}, $n = 10$

Interpolation: Position = $(30/100) \times 11 = 3.3 \rightarrow$ Between 3rd (6) and 4th (8) values

$$P_{30} = 6 + 0.3 \times (8 - 6) = 6 + 0.6 = \mathbf{6.6}$$

2 • Box Plot — Deep Dive & Variations

This session extends the box plot concepts from Part 1 with practical applications and comparison techniques.

2.1 Side-by-Side Box Plots

One of the most powerful uses of box plots: comparing the distribution of a **numerical variable across categories**.

Use Case

Comparing **salary distributions** across departments (Engineering, Marketing, Sales). Each department gets its own box, plotted side-by-side on the same axis. You can instantly compare medians, spreads, and outliers.

2.2 What You Can Read from Side-by-Side Box Plots

OBSERVATION	INTERPRETATION
Median lines at different heights	The groups have different central values
One box much wider (taller) than others	That group has more variability
Boxes don't overlap at all	Groups are likely significantly different
Boxes overlap heavily	Groups may not be significantly different
One group has many outlier dots	That group has more extreme values

2.3 Modified Box Plot (Tukey's Box Plot)

The standard box plot uses the **1.5×IQR rule** for whisker limits and outlier detection. Tukey also defined:

- **Mild outliers:** Between $1.5 \times \text{IQR}$ and $3 \times \text{IQR}$ from Q1/Q3
- **Extreme outliers:** Beyond $3 \times \text{IQR}$ from Q1/Q3

OUTLIER CLASSIFICATION

Mild Outlier: $Q1 - 3 \cdot \text{IQR} < x < Q1 - 1.5 \cdot \text{IQR}$ OR $Q3 + 1.5 \cdot \text{IQR} < x < Q3 + 3 \cdot \text{IQR}$

Extreme Outlier: $x < Q1 - 3 \cdot \text{IQR}$ OR $x > Q3 + 3 \cdot \text{IQR}$

3 • Introduction to Bivariate Analysis

While **univariate analysis** examines one variable at a time, **bivariate analysis** explores the relationship between **two variables simultaneously**. This is critical in ML for understanding feature interactions and dependencies.

3.1 The Three Combinations

VARIABLE 1	VARIABLE 2	KEY TECHNIQUES	KEY VISUALIZATIONS
Numerical	Numerical	Covariance, Correlation	Scatter Plot, Line Plot
Numerical	Categorical	Group statistics, ANOVA	Box Plot, Violin Plot, Bar Plot
Categorical	Categorical	Contingency Table, Chi-Square	Grouped Bar Chart, Stacked Bar, Heatmap

 **Why Bivariate Analysis Matters in ML**

Feature relationships are the backbone of ML models. Understanding how features interact helps with: **Feature selection** (keep correlated features with target), **Multicollinearity detection** (avoid highly correlated features in linear models), **Feature engineering** (create interaction features), and **Model interpretability**.

4 • Scatter Plots — Exploring Numerical-Numerical Relationships

A **scatter plot** displays individual data points on a 2D plane, with each axis representing a different numerical variable. It's the primary tool for visually assessing the relationship between two continuous variables.

4.1 What Scatter Plots Reveal

PATTERN	INTERPRETATION	VISUAL
Points rise from left to right	Positive relationship — as X increases, Y increases	↗ upward trend
Points fall from left to right	Negative relationship — as X increases, Y decreases	↘ downward trend
Points form a tight band	Strong relationship	Narrow cloud
Points widely scattered	Weak relationship	Diffuse cloud
Random cloud, no pattern	No linear relationship	Uniform blob
Curved pattern	Non-linear relationship (correlation won't capture this!)	Parabola/curve

4.2 Adding Dimensions to Scatter Plots

- **Color (hue):** Encode a third categorical variable using colour
- **Size:** Encode a fourth numerical variable using point size (bubble chart)
- **Shape:** Use different markers for categories
- **Regression line:** Overlay a best-fit line to show the trend

💡 Pro Tip

Always look at the **scatter plot before computing correlation**. Correlation only captures **linear** relationships. A strong curved relationship will show $r \approx 0$, which is misleading. The scatter plot will reveal the true pattern.

5 • Covariance — Direction of Joint Variability

Covariance measures how two variables **change together**. It tells you the **direction** of the linear relationship between two variables.

5.1 The Formula

FORMULA — COVARIANCE

Population: $\text{Cov}(X, Y) = \frac{\sum [(X_i - \mu_x)(Y_i - \mu_y)]}{N}$ Sample: $\text{Cov}(X, Y) = \frac{\sum [(X_i - \bar{X})(Y_i - \bar{Y})]}{(n - 1)}$

5.2 Intuition Behind the Formula

For each data point, we compute the product of its deviations from both means:

How the Sign of Covariance is Determined

$\bar{Y}$	
Quadrant II	Quadrant I
$(X_i - \bar{X}) < 0$	$(X_i - \bar{X}) > 0$
$(Y_i - \bar{Y}) > 0$	$(Y_i - \bar{Y}) > 0$
Product: $(-)(+)= -$	Product: $(+)(+)= +$
$\bar{X}$	
Quadrant III	Quadrant IV
$(X_i - \bar{X}) < 0$	$(X_i - \bar{X}) > 0$
$(Y_i - \bar{Y}) < 0$	$(Y_i - \bar{Y}) < 0$
Product: $(-)(-)= +$	Product: $(+)(-)= -$

Positive Cov → Most points in Quadrants I & III (same direction)

Negative Cov → Most points in Quadrants II & IV (opposite direction)

5.3 Interpretation

VALUE	MEANING	EXAMPLE
$\text{Cov}(X, Y) > 0$	X and Y move in the same direction	Study hours $\uparrow$ $\rightarrow$ Exam score $\uparrow$
$\text{Cov}(X, Y) < 0$	X and Y move in opposite directions	Price $\uparrow$ $\rightarrow$ Demand $\downarrow$
$\text{Cov}(X, Y) = 0$	No linear relationship	Shoe size & IQ

5.4 Special Property — Covariance with Itself

IMPORTANT PROPERTY

$\text{Cov}(X, X) = \sum [(X_i - \bar{X})(X_i - \bar{X})] / (n-1) = \sum [(X_i - \bar{X})^2] / (n-1) = \text{Var}(X)$ The covariance of a variable with itself equals its VARIANCE!

5.5 Limitations of Covariance

⚠ Critical Limitation

Covariance is **NOT standardised**. Its magnitude depends on the **units and scale** of the variables. A covariance of 1000 might be "small" for one pair and "large" for another. You cannot compare covariances across different variable pairs or datasets. **Solution** $\rightarrow$ **Use Correlation instead**.

📝 Step-by-Step Example

Data: $X = \{1, 2, 3, 4, 5\}$, $Y = \{2, 4, 5, 4, 5\}$

$\bar{X} = 3$, $\bar{Y} = 4$

Deviations & Products:

$$(1-3)(2-4) = (-2)(-2) = 4$$

$$(2-3)(4-4) = (-1)(0) = 0$$

$$(3-3)(5-4) = (0)(1) = 0$$

$$(4-3)(4-4) = (1)(0) = 0$$

$$(5-3)(5-4) = (2)(1) = 2$$

$$\sum \text{products} = 4 + 0 + 0 + 0 + 2 = 6$$

$$\text{Cov}(X, Y) = 6 / (5-1) = 6/4 = 1.5 \text{ (positive } \rightarrow \text{ same direction } \checkmark)$$

6 • Pearson Correlation Coefficient (r)

The **Pearson Correlation Coefficient (r)** solves the limitations of covariance by **normalising** it — producing a standardised, unit-free measure of the **strength and direction** of the linear relationship.

6.1 The Formula

FORMULA — PEARSON CORRELATION COEFFICIENT

$$r = \text{Cov}(X, Y) / (\sigma_x \cdot \sigma_y) \quad r = [\Sigma(X_i - \bar{X})(Y_i - \bar{Y})] / \sqrt{[\Sigma(X_i - \bar{X})^2 \cdot \Sigma(Y_i - \bar{Y})^2]}$$

6.2 The Range: -1 to +1

VALUE OF R	STRENGTH	DIRECTION	INTERPRETATION
r = +1	Perfect	Positive	All points lie on an upward-sloping line
+0.7 to +1	Strong	Positive	Clear upward trend with tight clustering
+0.4 to +0.7	Moderate	Positive	Noticeable upward trend but with scatter
+0.1 to +0.4	Weak	Positive	Slight upward trend, lots of scatter
r = 0	None	—	No linear relationship whatsoever
-0.1 to -0.4	Weak	Negative	Slight downward trend
-0.4 to -0.7	Moderate	Negative	Noticeable downward trend
-0.7 to -1	Strong	Negative	Clear downward trend
r = -1	Perfect	Negative	All points lie on a downward-sloping line

6.3 Properties of Pearson's r

- **Unit-free:** Doesn't depend on the scale or units of measurement
- **Bounded:** Always between -1 and +1
- **Symmetric:** $r(X, Y) = r(Y, X)$
- **Not affected by linear transformations:** Multiplying X by a constant or adding a constant doesn't change $|r|$
- **Only captures LINEAR relationships:** A parabolic relationship can have $r = 0$

6.4 Covariance vs Correlation — Side by Side

FEATURE	COVARIANCE	CORRELATION (R)
Range	$-\infty$ to $+\infty$	-1 to +1
Units	Product of units of X and Y	Unit-free (dimensionless)
Tells Direction?	✓ Yes	✓ Yes
Tells Strength?	✗ No (scale-dependent)	✓ Yes
Comparable?	✗ Cannot compare across datasets	✓ Directly comparable
Formula	$\sum (X_i - \bar{X})(Y_i - \bar{Y}) / (n-1)$	$\text{Cov}(X, Y) / (\sigma_X \cdot \sigma_Y)$

Continuing Previous Example

$\text{Cov}(X, Y) = 1.5$, $\sigma_X = \sqrt{2.5} \approx 1.581$, $\sigma_Y = \sqrt{1.5} \approx 1.225$

$r = 1.5 / (1.581 \times 1.225) = 1.5 / 1.937 \approx 0.775$

Interpretation: **Strong positive** linear relationship

7 • Correlation vs Causation

One of the most important concepts in all of statistics and data science: "**Correlation does NOT imply Causation.**"

⚠ The Golden Rule

Just because two variables are correlated (move together) does **NOT** mean one *causes* the other. This is one of the most common mistakes in data interpretation, journalism, and even published research.

7.1 Three Possible Explanations for Correlation

1. Direct Causation

X actually does cause Y.

Example: Smoking → Lung cancer
(Established through controlled experiments and decades of evidence)

2. Reverse Causation

Y actually causes X (not the other way around).

Example: "Countries with more hospitals have more deaths" — more hospitals are *built because* of higher population (more people → more hospitals AND more deaths)

3. Confounding Variable

A **hidden third variable Z** causes both X and Y, creating a false appearance of X → Y relationship.

Example: Ice cream sales ↑ and drowning deaths ↑ (both caused by hot weather, the confounder)

4. Pure Coincidence

Two variables happen to correlate by **random chance**, especially with many variables tested.

Example: Nicolas Cage movies vs swimming pool drownings (spurious correlation)

7.2 How to Establish Causation

- **Randomised Controlled Trials (RCTs):** The gold standard — randomly assign subjects to treatment/control groups
- **Natural Experiments:** Exploit random variation in real-world events
- **Temporal precedence:** Cause must come *before* the effect
- **Dose-response relationship:** More of the cause → more of the effect
- **Control for confounders:** Use multivariate analysis to hold other variables constant

8 • Spurious Correlations & Confounding Variables

8.1 Spurious Correlation

A **spurious correlation** is a statistical relationship between two variables that appears meaningful but is actually caused by coincidence or a hidden confounding variable.

Famous Spurious Correlations

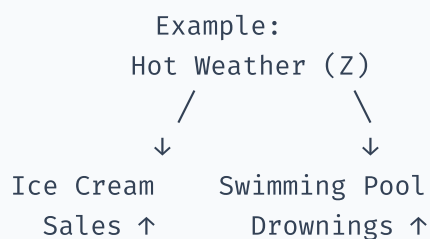
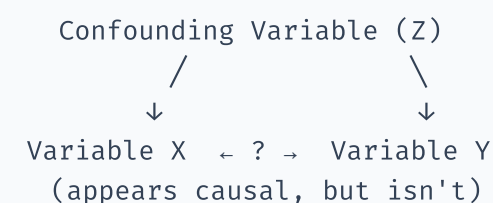
- **Per capita cheese consumption** correlates with **people dying tangled in bedsheets** ($r \approx 0.95$)
- **Number of Nicolas Cage films** correlates with **pool drowning deaths** ($r \approx 0.67$)
- **Divorce rate in Maine** correlates with **margarine consumption** ($r \approx 0.99$)

These are completely coincidental! With thousands of variables, some are bound to correlate by chance.

8.2 Confounding Variables (Lurking Variables)

A **confounding variable** is a hidden third variable that influences *both* the independent and dependent variables, making it appear that they are directly related.

Confounding Variable Structure



 How to Handle Confounders in ML

- **Include confounders as features** in your model so the model can account for them
- **Stratified analysis:** Analyse subgroups separately
- **Domain knowledge:** Consult subject matter experts to identify potential confounders
- **Multivariate regression:** Hold confounders constant while examining the relationship of interest

9 • Simpson's Paradox

Simpson's Paradox is a phenomenon where a trend that appears in **each subgroup of data** *reverses* or *disappears* when the subgroups are combined. It is one of the most counterintuitive results in statistics.

⚠ Why It's Dangerous

If you only look at **aggregate data** without examining subgroups, you can draw the **exact opposite conclusion** from what's actually true. This can lead to wrong decisions in healthcare, policy-making, ML model evaluation, and business.

9.1 Classic Example — UC Berkeley Admissions

📝 The Famous Example

Aggregate data: Overall, men had a higher admission rate than women → appears discriminatory.

Department-level data: When examined department by department, women actually had **equal or higher** admission rates in most departments!

Explanation: Women applied disproportionately to **more competitive departments** (lower acceptance rates), while men applied more to **easier-to-enter departments**. The *confounding variable* was the **choice of department**.

9.2 How to Detect Simpson's Paradox

- **Always stratify:** Break data down by relevant categorical variables
- **Compare aggregate vs subgroup trends:** If they disagree, investigate
- **Use Seaborn's `hue` parameter:** In pair plots and scatter plots, colour by categories to visually detect reversal
- **Domain knowledge:** Ask "what confounders could be driving this?"

💡 ML Lesson

When evaluating model performance, **always check metrics at the subgroup level** (by gender, age group, etc.) — not just aggregate accuracy. A model might perform well overall but fail on specific subgroups (fairness problem).

10 • Bivariate Analysis — Numerical vs Categorical

When one variable is **numerical** and the other is **categorical**, the goal is to see how the numerical variable's distribution differs across categories.

10.1 Key Techniques

Side-by-Side Box Plots

The most effective visualization — shows median, spread, quartiles, and outliers for each category.

Violin Plots

Combines a box plot with a KDE (kernel density estimate). Shows the full distribution shape, not just summary statistics. Wider sections indicate more data points at that value.

Bar Plot with Error Bars

Shows the **mean** of the numerical variable per category with confidence interval error bars. Useful for quick comparisons but **hides the distribution shape**.

Dist Plots / Histograms per Category

Overlay histograms or KDE curves for each category to compare full distributions.

10.2 Statistical Summary per Group

Example — Salary by Department

Use `df.groupby('department')['salary'].describe()` to get a complete statistical summary (count, mean, std, min, 25%, 50%, 75%, max) for each department. Then visualise with box plots to confirm the patterns.

11 · Bivariate Analysis — Categorical vs Categorical

When both variables are **categorical**, we examine how the frequency distribution of one variable changes across the levels of the other.

11.1 Contingency Table (Cross-tabulation)

A **contingency table** (crosstab) is a matrix that displays the frequency distribution of two categorical variables simultaneously.

Example — Gender × Survival on Titanic

	SURVIVED	DID NOT SURVIVE	TOTAL
Female	233	81	314
Male	109	468	577
Total	342	549	891

This immediately reveals: 74% of females survived vs only 19% of males — a strong relationship between Gender and Survival.

11.2 Visualizations for Categorical × Categorical

Grouped (Clustered) Bar Chart

Bars for each combination plotted side by side. Best for comparing frequencies across groups.

Stacked Bar Chart

Bars stacked on top of each other. Shows both individual and total frequencies. Good for seeing proportions.

Heatmap of the Crosstab

Colour-coded matrix showing the crosstab values. Darker colours = higher frequencies. Best for large number of categories.

12 · Multivariate Visualization — Pair Plots, Heatmaps & Beyond

When you need to examine relationships among **three or more variables** simultaneously, you need specialised visualisation tools.

12.1 Pair Plot (Seaborn `pairplot`)

Creates a **matrix of scatter plots** for every pair of numerical variables. The diagonal shows the distribution of each individual variable (histogram or KDE).

What It Reveals

- **All pairwise relationships** at a glance
- **Clusters** when coloured by a categorical variable (hue)
- **Outliers** visible in individual scatter panels
- **Distribution shapes** on the diagonal
- **Simpson's paradox** — trends may reverse when hue is applied

💡 Key Parameters

`hue` — colour by a categorical variable (essential for subgroup analysis)

`vars` — select specific columns to plot

`diag_kind` — 'hist' or 'kde' for diagonal plots

`kind` — 'scatter' or 'reg' (adds regression lines)

12.2 Correlation Heatmap

A **colour-coded matrix** showing the Pearson correlation coefficient between every pair of numerical variables.

Reading a Heatmap

- **Dark red/positive:** Strong positive correlation
- **Dark blue/negative:** Strong negative correlation
- **White/neutral:** No correlation
- **Diagonal:** Always 1 (variable correlated with itself)
- **Symmetric:** $r(X,Y) = r(Y,X)$, so the matrix is symmetric about the diagonal

💡 ML Use Case — Feature Selection

Use the correlation heatmap to:

- **Find features correlated with the target** (keep them)
- **Find highly correlated features with each other** (remove one to avoid multicollinearity)
- A rule of thumb: features with $|r| > 0.85$ between them often cause problems in linear models

12.3 Other Multi-Variable Techniques

TECHNIQUE	BEST FOR	LIBRARY
Pair Plot	All pairwise scatter plots + distributions	<code>sns.pairplot()</code>
Correlation Heatmap	Correlation matrix visualisation	<code>sns.heatmap(df.corr())</code>
Parallel Coordinates	Comparing patterns across many features	<code>pd.plotting.parallel_coordinates()</code>
3D Scatter Plot	Three numerical variables	<code>plt.axes(projection='3d')</code>
Facet Grid	Repeated plots for subsets	<code>sns.FacetGrid()</code>

13 · Python Implementation — Complete Code

13.1 Covariance & Correlation

```
import numpy as np
import pandas as pd

# Sample data
df = pd.DataFrame({
    'study_hours': [1, 2, 3, 4, 5, 6, 7, 8],
    'exam_score': [40, 45, 55, 60, 70, 75, 85, 90],
    'sleep_hours': [9, 8, 7, 7, 6, 6, 5, 5]
})

# — Covariance Matrix —
print("Covariance Matrix:")
print(df.cov())

# — Correlation Matrix (Pearson) —
print("\nCorrelation Matrix:")
print(df.corr()) # default: method='pearson'

# — Individual correlation —
r = df['study_hours'].corr(df['exam_score'])
print(f"\nr(study_hours, exam_score) = {r:.4f}")

# — Using NumPy —
cov_matrix = np.cov(df['study_hours'], df['exam_score'])
corr_matrix = np.corrcoef(df['study_hours'], df['exam_score'])
```

13.2 Scatter Plot with Regression Line

```
import matplotlib.pyplot as plt
import seaborn as sns

# Basic scatter plot
plt.figure(figsize=(8, 5))
sns.scatterplot(data=df, x='study_hours', y='exam_score',
               s=100, color='#3b82f6', edgecolor='white')

# Add regression line
sns.regplot(data=df, x='study_hours', y='exam_score',
            scatter=False, color='#ef4444', line_kws={'linewidth': 2})

plt.title('Study Hours vs Exam Score', fontsize=14, fontweight='bold')
plt.xlabel('Study Hours')
plt.ylabel('Exam Score')
plt.tight_layout()
plt.show()
```

13.3 Side-by-Side Box Plot & Violin Plot

```
# Using Titanic dataset
titanic = sns.load_dataset('titanic')

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Side-by-side Box Plot
sns.boxplot(data=titanic, x='class', y='age', palette='viridis', ax=axes[0])
axes[0].set_title('Age Distribution by Passenger Class')

# Violin Plot
sns.violinplot(data=titanic, x='class', y='age', palette='muted', ax=axes[1])
axes[1].set_title('Age Distribution (Violin) by Class')

plt.tight_layout()
plt.show()
```

13.4 Contingency Table & Grouped Bar Chart

```
# Contingency Table (Crosstab)
ct = pd.crosstab(titanic['sex'], titanic['survived'], margins=True)
print(ct)

# Grouped Bar Chart
ct_no_margins = pd.crosstab(titanic['sex'], titanic['survived'])
ct_no_margins.plot(kind='bar', figsize=(8, 5), color=['#ef4444', '#22c55e'])
plt.title('Survival by Gender')
plt.xlabel('Gender')
plt.ylabel('Count')
plt.xticks(rotation=0)
plt.legend(['Did Not Survive', 'Survived'])
plt.tight_layout()
plt.show()
```

13.5 Pair Plot & Correlation Heatmap

```
# — Pair Plot —
iris = sns.load_dataset('iris')
sns.pairplot(iris, hue='species', diag_kind='kde', palette='Set2')
plt.suptitle('Iris Dataset — Pair Plot', y=1.02, fontsize=14)
plt.show()

# — Correlation Heatmap —
plt.figure(figsize=(8, 6))
corr = iris.drop('species', axis=1).corr()
sns.heatmap(corr, annot=True, cmap='coolwarm', center=0,
            fmt='.2f', square=True, linewidths=1,
            vmin=-1, vmax=1)
plt.title('Correlation Heatmap — Iris Features', fontsize=14, fontweight='bold')
plt.tight_layout()
plt.show()
```

14 · Quick Reference Card

Descriptive Statistics Part 2 — Complete Cheat Sheet

Bivariate Analysis Summary

- **Num × Num** → Scatter plot, Covariance, Correlation
- **Num × Cat** → Side-by-side Box/Violin plot, Group statistics
- **Cat × Cat** → Contingency table (crosstab), Grouped bar chart, Heatmap

Covariance & Correlation

- $\text{Cov}(X, Y) = \sum (X_i - \bar{X})(Y_i - \bar{Y}) / (n-1) \rightarrow$ direction only, unbounded
- $r = \text{Cov}(X, Y) / (\sigma_X \cdot \sigma_Y) \rightarrow$ direction + strength, bounded $[-1, +1]$
- $\text{Cov}(X, X) = \text{Var}(X)$ — covariance of a variable with itself = its variance

Correlation Strength Guide

- $|r| > 0.7 \rightarrow$ **Strong**
- $0.4 < |r| < 0.7 \rightarrow$ **Moderate**
- $|r| < 0.4 \rightarrow$ **Weak**
- $r = 0 \rightarrow$ **No linear relationship** (may still be non-linear!)

Critical Warnings

-  **Correlation ≠ Causation** — always consider confounders
-  **Pearson r captures only LINEAR relationships**
-  **Simpson's Paradox** — aggregate trends can reverse at subgroup level
-  **Always plot first, compute later**

Python Quick Reference

- `df.corr()` — Full correlation matrix
- `df.cov()` — Full covariance matrix
- `pd.crosstab(df['a'], df['b'])` — Contingency table
- `sns.pairplot(df, hue='col')` — Pair plot with categories
- `sns.heatmap(df.corr(), annot=True)` — Correlation heatmap
- `sns.boxplot(x='cat', y='num', data=df)` — Side-by-side box plot
- `sns.violinplot(x='cat', y='num', data=df)` — Violin plot

 What's Next? → Probability Distribution Functions (Session 40)

Part 3 of the series moves into **Probability Distribution Functions** — covering PDFs, PMFs, CDFs, and the mathematical foundations of probability that underpin all of inferential statistics and machine learning.